

SS1 Mathematics Diagnostic: Cambridge IGCSE Standard (Part 2)

Scope: Advanced Algebra, Indices, and Set Logic
Target: All SS1 Students (Questions 21 - 40)

Section A: Advanced Algebra & Indices

No.	Question & Options	Ans	Diagnostic Focus
21	Solve the simultaneous equations: $3x + y = 7$ and $x - 2y = 0$ A) $x=2, y=1$ B) $x=1, y=2$ C) $x=3, y=-2$ D) $x=0, y=7$	A	Algebraic Systems (Intersection).
22	Simplify: $\frac{2a^2b \times 3ab^4}{6a^3b^2}$ A) $a b^3$ B) b^3 C) $a^2 b^3$ D) 1	B	Laws of Indices (Simplification).
23	Find the value of x if $2^{x-1} = 32$ A) 4 B) 5 C) 6 D) 7	C	Exponential Equations.
24	Express 0.0000452 in standard form (scientific notation). A) 4.52×10^{-5} B) 4.52×10^5 C) 45.2×10^{-6} D) 0.452×10^{-4}	A	Standard Form & Precision.
25	Factorize: $x^2 - 5x + 6$ A) $(x-1)(x-6)$ B) $(x-2)(x-3)$ C) $(x+2)(x+3)$ D) $(x-2)(x+3)$	B	Quadratic Expressions (Middle term splitting).
26	Identify the range of x for the inequality: $3x - 5 > 10$. A) $x > 5$ B) $x < 5$ C) $x > 15$ D) $x \geq 5$	A	Linear Inequalities.
27	A map has a scale of $\$1 : 200,000$. If two towns are 5 cm apart on the map, what is the actual distance in kilometers? A) 1 km B) 10 km C) 100 km D) 1,000 km	B	Ratio and Scale (Cambridge IGCSE skill).
28	Calculate the perimeter of a semi-circle with radius 7 cm. (Take $\pi = \frac{22}{7}$) A) 22 cm B) 36 cm C) 44 cm D) 58 cm	B	Mensuration (Confusion check: Arc vs Perimeter).
29	Solve for p : $\sqrt{2p + 1} = 5$ A) 2 B) 10 C) 12 D) 13	C	Radical Equations.
30	Which of the following is equivalent to $\log_2 8 + \log_2 4$? A) 5 B) 6 C) 12 D) 32	A	Logarithm Laws (Product rule).

Section B: Advanced Set Theory & Sequences

No.	Question & Options	Ans	Diagnostic Focus
31	The n -th term of a sequence is given by $U_n = 2n^2 - 1$. Find the 4th term. A) 15 B) 31 C) 63 D) 7	B	Sequences & Series foundations.
32	Find the next term in the sequence: 1, 4, 9, 16, ... A) 20 B) 25 C) 36 D) 49	B	Pattern Recognition (Square numbers).
33	In a class of 30 students, 18 play football, 15 play basketball, and 2 play neither. How many play both? A) 3 B) 5 C) 10 D) 12	B	Advanced Venn Diagram Application.
34	Evaluate: $27^{-\frac{2}{3}}$ A) 9 B) $\frac{1}{9}$ C) -9 D) $\frac{1}{3}$	B	Negative & Fractional Indices.

No.	Question & Options	Ans	Diagnostic Focus
35	If y is inversely proportional to x , and $y=4$ when $x=3$, find y when $x=6$. A) 2 B) 8 C) 12 D) 1.5	A	Variation (Inverse).
36	Calculate the total surface area of a cube with side length 4 cm. A) 64 cm^2 B) 96 cm^2 C) 16 cm^2 D) 48 cm^2	B	Space Geometry (Area).
37	Expand and simplify: $(2x - 3)(x + 4)$ A) $2x^2 + 5x - 12$ B) $2x^2 - 5x + 12$ C) $2x^2 + 8x - 12$ D) $2x^2 + x - 12$	A	Polynomial Expansion.
38	Evaluate $\log_3 1$. A) 1 B) 3 C) 0 D) Undefined	C	Logarithm Identity.
39	Solve $x^2 - 9 = 0$. A) $x=3$ B) $x=9$ C) $x=3$ or $x=-3$ D) $x=-3$	C	Quadratic Solutions (Two roots).
40	Diagnostic Insight: Students failing Q21 and Q25 are not ready for SS1 Science/Commercial core; need remediation in Algebra.	-	Retention analysis.